Lab 2 – AI² Product Specification

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the AI² (Artificially Intelligent Invoices) system. It serves as a formal agreement on the software's capabilities, constraints, and design expectations so as to ensure consistency in development and clear communication between stakeholders. The intended audience includes the development team, reviewers, ADS stakeholders, CS411W instructor, and any personnel responsible for reviewing, validating, or implementing the AI² prototype.

1.2 Scope

The AI² system is designed to be an intelligent automation system that streamlines invoice intake, organization, and purchase order matching. AI² will automatically retrieve invoices from an Outlook GCC High email inbox, extract and parse invoice data, match invoices to purchase orders in Oracle NetSuite, assign confidence scores, automatically process invoices that meet or surpass a set confidence score, route exceptions and invoices with low confidence scores for human review, and maintain full audit traceability.

The system will not execute any payments, perform vendor onboarding, or replace the full functionality of an ERP system. Additionally, AI² will not instill a prioritization system for emails that are not invoice-related. It is intended to reduce manual workload, accelerate payment readiness, and improve accuracy for organizations with high-volume Accounts Payable operations. The goals of the system include increasing invoice processing efficiency, reduction of errors, and enhancing compliance visibility.

1.3 Definitions, acronyms, and abbreviations

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

Aperture: A spreadsheet extraction tool for Excel-based invoices.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

OCR (Optical Character Recognition): This will be used to extract text from documents.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Textract: An AWS OCR service for PDF parsing.

Vendor: A person or company offering goods or services for sale.

1.4 References

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1.5 Overview

The remaining sections of this Software Requirements Specification (SRS) document will provide a detailed breakdown of the AI² system. The sections following this section will go over AI²'s overall description, specific functional and non-functional requirements, design constraints, and others as will be seen in the next few sections. The sections together will define the complete set of requirements for the AI² prototype.

2. Overall Description

2.1 Product Perspective

AI² is a component of a larger Accounts Payable ecosystem that includes Outlook GCC High, Oracle, NetSuite, Alteryx, AWS Textract, Aperture, and internal ADS authentication services. While the prototype will simulate these integrations, the full system will operate as an in interconnected automation platform that retrieves invoices, extracts data, performs PO matching, assigns confidence scores, and routes exceptions for review. AI² fits into the broader AP workflow as the intelligent front-end processor that prepares invoices for payment readiness. In addition to this, AI² operates within several constraints to include secure communication with Microsoft Graph API, Oracle NetSuite via Alteryx, and AWS GovCloud services. The system must comply with ADS's Okta-based authentication and government compliance frameworks to include CMMC and FedRAMP.

To start, the system interfaces for this system's future implementation include Outlook GCC High, Oracle NetSuite, AWS Textract, and Aperture. Outlook GCC High is the email inbox in which all the invoices will be located in from which the system will need to retrieve invoice emails and attachments via the Microsoft Graph API. Oracle NetSuite is where the list of PO numbers will be located in from which the system will query PO, vendor, and line-item data through the Alteryx. AWS Textract and Aperture will be responsible for extracting text from PDFs and spreadsheets. However, it must be noted that the prototype will only simulate these interfaces using files provided by ADS and mock data.

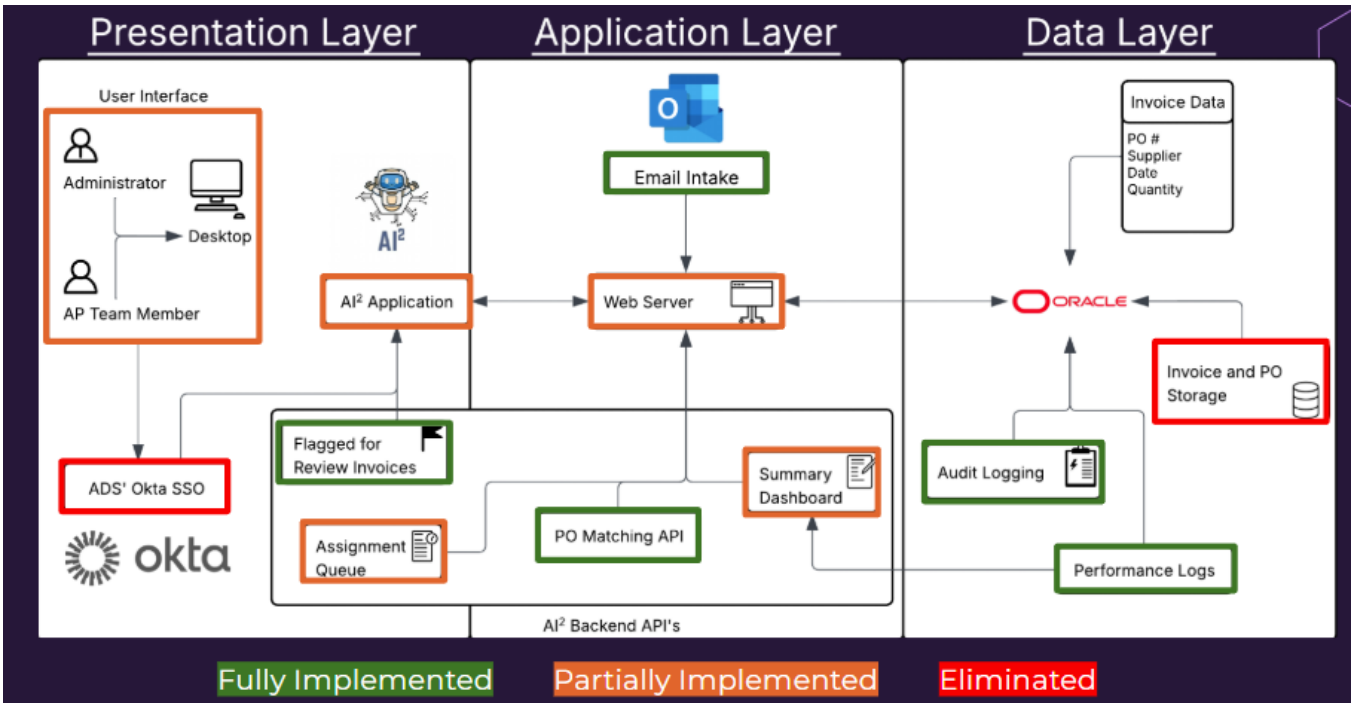
Next, the user interface in which AI² provides a web-based dashboard that will provide the user with the option to upload an invoice in the prototype, confidence score display, PO match results, exception review queue, supplier-based routing, and audit logging visibility. However, in

future implementations, the system will automatically retrieve invoices directly from the email inbox instead of needing to upload invoices. The interface will be designed for clarity, minimal training time of the model, and accessibility for AP staff with varying technical backgrounds.

Moving on to the hardware interfaces in which the full system will be hosted in AWS GovCloud that will be accessible via standard office workstations. However, for the prototype, AI² will be hosted locally. For the software interfaces, the full system will have an outlook integration via the Microsoft Graph API, an Oracle NetSuite data exchange via the Alteryx API, PDF and spreadsheet parsing will be done through AWS Textract and Aperture, and PostgreSQL will be used to structure invoice and audit log storage. However, the prototype will only use mock data and local storage instead of external APIs. The architecture of the AI² prototype will have three layers to include the presentation layer, an application layer, and a data layer that is shown in Figure 1 below.

Figure 1

AI² Prototype MFCD



2.2 Product Functions

AI² retrieves invoices from Outlook in the full implementation or can accept manual uploads in the prototype. Additionally, the system will parse PDFs and spreadsheets to extract invoice fields as well as match invoices to purchase orders, line items, and amounts. The system will also assign confidence levels and flag any low-confidence matches and missing data for human review to AP staff. Any corrections made will be stored to improve future accuracy and invoices will be automatically assigned to AP representatives. Furthermore, all system and user actions will be recorded for compliance. A more detailed breakdown of what the full implementation vs the prototype will have are demonstrated in Table 1 below.

Table 1

AI² Real World Product vs Prototype

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Outlook Email Integration (Graph API)	Fully Functional	Partially Functional	Simulate Intake via PDFs and local folder ingestion
Invoice Parser and Data Extractor	Fully Functional	Fully Functional	Show Progress On Various Formats over the course of the semester
Automated PO Matching Algorithm	Fully Functional	Fully Functional	
Invoice Quality Analyzer	Fully Functional	Fully Functional	
Queue Assigning and Routing	Fully Functional	Partially Functional	Start with a simple container, build from there
Queue Assignment Dashboard	Fully Functional	Partially Functional	Time constraint, Build a Minimal Viable Dashboard

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Admin Configuration Dashboard	Fully Functional	Partially Functional	Confidence Threshold, Queue routing
System Health and Performance Dashboard	Fully Functional	Partially Functional	Core elements like, error count, matching success, invoice processed, etc.
Error Logging and Recovery Module	Fully Functional	Fully Functional	
Admin System Alert Notifications	Fully Functional	Partially Functional	Push notifications

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Automated Supplier Notifications	Fully Functional	Eliminated	No external comms
Automated AP Team Reminders	Fully Functional	Partially Functional	Push notification
AI Confidence Scoring Module	Fully Functional	Fully Functional	
Continuous Learning From Human Interaction	Fully Functional	Fully Functional	
Human Review Panel	Fully Functional	Fully Functional	

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Encryption Verification Tool	Partially Functional	Eliminated, Out of Scope	Most encryption handled by external systems (AWS)
AWS GovCloud Integration Tool	Fully Functional	Eliminated	Outside the scope of one semester
Audit Log Management System	Fully Functional	Partially Functional	Simulated, Basic logic to DB or file
Compliance Changelog System	Fully Functional	Partially Functional	Changes tracked manually
Auto Generated Compliance Report	Fully Functional	Eliminated	Mocked output, sample JSON
Compliance Regulations Monitoring Tool	Fully Functional	Eliminated	Out of Scope

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Okta SSO Authentication Module	Fully Functional	Eliminated	Specific to ADS, Would require their authorization
Alteryx-Oracle Integration Module	Fully Functional	Eliminated	Specific to ADS, Would require their authorization
Integration Monitoring Dashboard	Fully Functional	Eliminated	Specific to ADS, Would require their authorization

2.3 User Characteristics

AI² is designed for the user roles of AP representatives, AP team leads, and administrators. AP representatives would have moderate computer efficiency and be responsible for reviewing invoices, correcting mismatches, and approving of matches. AP team leads would be experienced users who monitor queues, reassign suppliers to AP representatives, and oversee exceptions. System administrators would be the technical staff responsible for the configuration of the system, user management, and system oversight. This system holds no requirement for programming or technical expertise as the above characteristics for each user role would require the development of a simple, intuitive interface and minimal training requirements.

2.4 Constraints

AI² has multiple constraints that would influence the system's design and implementation. For the regulatory constraints, AI² must meet the CMMC, FedRAMP, and ADS internal compliance requirements as the invoices being taken in contain sensitive, government data. Hardware constraints include the need for AWS GovCloud deployment in the full system and standard workstation access for end-users. Additionally, for the software constraints, this would involve

integration with Outlook, Oracle NetSuite, Alteryx, Textract, and Aperture. All of which are third-party services while also taking into consideration the requirement for a secure authentication through Okta SSO. In turn, reliability is critical since any delays or errors while invoices are being processed can impact supplier payments and compliance regulations and standards. Moreover, there are also security constraints to include ensuring that all communication is encrypted and that there is enforcement of strict access controls and audit logging. However, the prototype is constrained by the absence of real integrations and will rely on simulated data and simplified workflows.

2.5 Assumptions and Dependencies

It will be assumed that ADS will maintain access to Outlook GCC High, Oracle NetSuite, AWS services, and internal authentication systems. Additionally, it will be assumed that AP staff will continue to follow existing procurement workflows and that network connectivity and authentication services remain available. As for the prototype itself, dependencies will be on mock data provided by ADS, manually uploading invoices, and having simplified configurations to simulate in place of real integrations. Any changes to these assumptions and dependencies may require revisions to the system requirements.